Пункт 1. Вход под пользователем user1

```
root@eltex-practice2-pg1-v17:~# su - user1
user1@eltex-practice2-pg1-v17:~$
```

Пункт 2. Подсчёт процессов с несколькими потоками

```
user1@eltex-practice2-pg1-v17:~$ ps -eLf | awk '{print $2}' | sort | uniq -c | awk '$1 > 1' | wc -l
10
user1@eltex-practice2-pg1-v17:~$
```

Пункт 3: Запустите top и настройте вывод полей

Top

```
top - 13:07:22 up 12 days,  4:40,  1 user,  load average: 0.00, 0.00, 0.00
Tasks: 107 total,   1 running, 106 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 98.3 id,  1.7 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3916.0 total,   1143.2 free,    558.0 used,   2538.7 buff/cache
MiB Swap:   3185.0 total,   3185.0 free,      0.0 used.   3358.1 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   22472  13696   9472 S   0.0   0.3   0:22.59 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.11 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_g
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_p
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.22 kworker/0:0H-kblockd
     12 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_pe
     13 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.30 ksoftirqd/0
     17 root      20   0       0      0      0 I   0.0   0.0   0:09.47 rcu_preempt
     18 root      rt   0       0      0      0 S   0.0   0.0   0:03.98 migration/0
     19 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     21 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     22 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
     23 root      rt   0       0      0      0 S   0.0   0.0   0:04.13 migration/1
     24 root      20   0       0      0      0 S   0.0   0.0   0:00.29 ksoftirqd/1
     26 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
     27 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
```

Удалены поля VIRT, RES, SHR

Добавлено поле RUSER, и показано после поля USER

```
Fields Management for window 1:Def, whose current sort field is %CPU
   Navigate with Up/Dn, Right selects for move then <Enter> or Left commits,
   'd' or <Space> toggles display, 's' sets sort.  Use 'q' or <Esc> to end!

* PID      = Process Id            TIME     = CPU Time              RSan     = RES Anonymous (KiB)
* USER     = Effective User Name   SWAP     = Swapped Size (KiB)    RSfd     = RES File-based (KiB)
* RUSER    = Real User Name        CODE     = Code Size (KiB)       RSlk     = RES Locked (KiB)
* PR       = Priority              DATA     = Data+Stack (KiB)      RSsh     = RES Shared (KiB)
* NI       = Nice Value            nMaj     = Major Page Faults     CGNAME   = Control Group name
  VIRT     = Virtual Image (KiB)   nMin     = Minor Page Faults     NU       = Last Used NUMA node
  RES      = Resident Size (KiB)   nDRT     = Dirty Pages Count     LOGID    = Login User Id
  SHR      = Shared Memory (KiB)   WCHAN    = Sleeping in Function  EXE      = Executable Path
* S        = Process Status        Flags    = Task Flags <sched.h>  RSS      = Res Mem (smaps), KiB
* %CPU     = CPU Usage             CGROUPS  = Control Groups        PSS      = Proportion RSS,  KiB
* %MEM     = Memory Usage (RES)    SUPGIDS  = Supp Groups IDs       PSan     = Proportion Anon, KiB
* TIME+    = CPU Time, hundredths  SUPGRPS  = Supp Groups Names     PSfd     = Proportion File, KiB
* COMMAND  = Command Name/Line     TGID     = Thread Group Id       PSsh     = Proportion Shrd, KiB
  PPID     = Parent Process pid    OOMa     = OOMEM Adjustment      USS      = Unique RSS, KiB
  UID      = Effective User Id     OOMs     = OOMEM Score current   ioR      = I/O Bytes Read
  RUID     = Real User Id          ENVIRON  = Environment vars      ioRop    = I/O Read Operations
  SUID     = Saved User Id         vMj      = Major Faults delta    ioW      = I/O Bytes Written
  SUSER    = Saved User Name       vMn      = Minor Faults delta    ioWop    = I/O Write Operations
  GID      = Group Id              USED     = Res+Swap Size (KiB)   AGID     = Autogroup Identifier
  GROUP    = Group Name            nsIPC    = IPC namespace Inode   AGNI     = Autogroup Nice Value
  PGRP     = Process Group Id      nsMNT    = MNT namespace Inode   STARTED  = Start Time from boot
  TTY      = Controlling Tty       nsNET    = NET namespace Inode   ELAPSED  = Elapsed Running Time
  TPGID    = Tty Process Grp Id    nsPID    = PID namespace Inode   %CUU     = CPU Utilization
  SID      = Session Id            nsUSER   = USER namespace Inode  %CUC     = Utilization + child
  nTH      = Number of Threads     nsUTS    = UTS namespace Inode   nsCGROUP = CGRP namespace Inode
  P        = Last Used Cpu (SMP)   LXC      = LXC container name    nsTIME   = TIME namespace Inode
```

4.В другом терминальном окне выполните команду passwd и оставьте ее в состоянии запроса текущего пароля

```
root@eltex-practice2-pg1-v17:~# su - user1
user1@eltex-practice2-pg1-v17:~$ passwd
Changing password for user1.
Current password:
```

Пункт 5: В окне с top

Shift + u

User1

```
top - 13:13:15 up 12 days,  4:46,  2 users,  load average: 0.00, 0.00, 0.00
Tasks: 114 total,   1 running, 113 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3916.0 total,   1140.1 free,    561.1 used,   2538.8 buff/cache
MiB Swap:   3185.0 total,   3185.0 free,      0.0 used.   3355.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  82247 user1     20   0    8868   5632   3840 S   0.0   0.1   0:00.02 bash
  82350 user1     20   0    8644   5376   3840 S   0.0   0.1   0:00.01 bash
  82360 root      20   0    9172   3712   3456 S   0.0   0.1   0:00.00 passwd
  82361 user1     20   0   11912   5888   3712 R   0.0   0.1   0:00.00 top
```

Отправил процессу passwd сигналы 15 (SIGTERM), 2 (SIGINT), 3 (SIGQUIT), 9(SIGKILL)

```
top - 13:19:36 up 12 days,  4:52,  2 users,  load average: 0.00, 0.00, 0.00
Tasks: 112 total,   1 running, 111 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3916.0 total,   1142.2 free,    558.9 used,   2538.8 buff/cache
MiB Swap:   3185.0 total,   3185.0 free,      0.0 used.   3357.1 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  82247 user1     20   0    8868   5632   3840 S   0.0   0.1   0:00.02 bash
  82350 user1     20   0    8644   5376   3840 S   0.0   0.1   0:00.01 bash
  82369 user1     20   0   11912   6016   3840 R   0.0   0.2   0:00.11 top
```

6.Выполните команду vim ~/file_task3.txt и нажмите Ctrl-Z

```
"~/.file_task3.txt" [New]                                    0,0-1         All
```

```
user1@eltex-practice2-pg1-v17:~$ vim ~/.file_task3.txt

[1]+  Stopped                 vim ~/.file_task3.txt
[1]+  Stopped                 vim ~/.file_task3.txt
user1@eltex-practice2-pg1-v17:~$ jobs
[1]+  Stopped                 vim ~/.file_task3.txt
user1@eltex-practice2-pg1-v17:~$
```

7.Выполните команду sleep 600, нажмите Ctrl-Z и выполните команду jobs

```
user1@eltex-practice2-pg1-v17:~$ sleep 600
jobs
^Z
[2]+  Stopped                 sleep 600
user1@eltex-practice2-pg1-v17:~$ jobs
[1]-  Stopped                 vim ~/.file_task3.txt
[2]+  Stopped                 sleep 600
user1@eltex-practice2-pg1-v17:~$
```

8.Последнее задание (sleep 600) сделайте фоновым

```
user1@eltex-practice2-pg1-v17:~$ bg %2
[2]+ sleep 600 &
user1@eltex-practice2-pg1-v17:~$
```

9.Измените число NICE у задания (sleep 600), сделав его равным 10

```
user1@eltex-practice2-pg1-v17:~$ renice 10 -p 82393
82393 (process ID) old priority 0, new priority 10
user1@eltex-practice2-pg1-v17:~$
```

10 Проверьте, что число NICE у этого задания изменилось

```
user1@eltex-practice2-pg1-v17:~$ ps -l -p 82393
F S   UID     PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
0 S  1001   82393   82247  0  90  10 -  1421 do_sys pts/0    00:00:00 sleep
user1@eltex-practice2-pg1-v17:~$
```

11 Сделайте задание vim ~/file_ task3.txt активным и выйдите из редактора

```
user1@eltex-practice2-pg1-v17:~$ fg %1
vim ~/.file_task3.txt
user1@eltex-practice2-pg1-v17:~$
```

```
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:q!
```

12 Отправьте сигнал 15 (SIGTERM) заданию sleep 600 и выполните команду jobs

```
user1@eltex-practice2-pg1-v17:~$ jobs
[2]+  Terminated              sleep 600
user1@eltex-practice2-pg1-v17:~$
```

13.Создайте перехватчик сигналов SIGINT и SIGQUIT внутри командного интерпретатора, который выводит сообщение «Меня голыми руками не возьмёшь!» (используйте встроенную команду trap) и отправьте сигналы самому себе

```
user1@eltex-practice2-pg1-v17:~$ trap 'echo "Меня голыми руками не возьмёшь!"' INT QUIT
user1@eltex-practice2-pg1-v17:~$ kill -INT $$
Меня голыми руками не возьмёшь!
user1@eltex-practice2-pg1-v17:~$ kill -QUIT $$
Меня голыми руками не возьмёшь!
user1@eltex-practice2-pg1-v17:~$
```